

Met Gala 2026

Social Media and Networks Analysis (COSC3047)

Assignment 2: Undergraduate Level Report

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GitHub Repository: <https://github.com/gnol-123/Network-Analysis-Project>

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1. Problem Formation

1.1 Project Context

The Met Gala is an annual charity event that benefits the Costume Institute of the Metropolitan Museum of Art ('the Met') in New York (Britannica, 2026). The event each year is given a theme where celebrities represent different designer brands, dressed in extravagant, statement fashion pieces. The Met Gala features many high-profile celebrities, designers, and cultural figures, making it a display of power, class, status, and influence.

This analysis is performed on YouTube comment discussion of the 2026 Met Gala held on the 4/6/26 over a week long period, focused on which celebrities and designer brands were core to discussion. Analysis will focus on overall sentiment surrounding different entities, entities that are associated through co-mentions, and different topics that arise from analysing co-occurrence between different comments.

1.2 Research Questions

The proposed research questions being explored are:

- Which celebrities and designer brands dominated YouTube audience discussion of the 2026 Met Gala?
- How were these figures associated through comment co-mentions?
- Are there identifiable communities of related celebrities and designer brand co-mentions?
- What sentiment patterns and topics emerged around the most discussed celebrities, designers, outfits, and collaborations?

1.3 Justification

The selected research questions are interesting to investigate as the Met Gala is subject to many different angles of criticism and praise. Fans approach the Met Gala from an artistic perspective celebrating beauty and fashion or as fans of celebrities, whereas critics generally focus on the display of wealth, power, and influence.

YouTube is an appropriate data source as the Met Gala is streamed and results in commentary videos, fashion reviews, theme reviews, and reaction videos providing a broad range of perspective. Assessing associated comments will provide a range of different sentiment and network analysis.

A network between celebrities and designer brands will be formed with each of these entities representing the nodes in the network. Co-mentions for celeb-celeb or celeb-brand connections will form the edges in this graph with counts on these co-mention relationships representing weights. The formation of this network graph will allow for clusters of related entities to be identified.

Investigation of a luxurious, opulent event at this point in time is expected to generate strong opinions on either side given growing discussions around class inequality and political frustrations.

2. Data Collection and Limitations

2.1 Data Source

This project investigates sentiment surrounding the 2026 Met Gala by performing analysis on comments retrieved via the YouTube data API.

The main aim is to identify:

- the celebrities and brands mentioned most frequently
- the sentiment expressed toward these celebrities and brands
- common co-mentions between celebrities and designer brands
- network groupings that emerge between different celebrities and designer brands

YouTube data has been selected as the Met Gala generates videos in the form of streams, commentaries, reactions, critiques with relevant discussion occurring in the comments where users discuss the attending celebrities and designer brands. By analysing comments attached to these videos, evidence-backed sentiment on the event and its various entities can be determined, providing a grounded assessment on how the general public perceives the event in addition to different celebrities and brands.

The project will construct a co-mention network using celebrities and designer brands as nodes. Edges will be constructed through co-mentions of different combinations of celebrities and designer brands within individual comments and comment threads (e.g. a celebrity and the brand they are representing). The subsequent network analysis will be used to determine primary celebrities and brands discussed, celebrity to celebrity relations, brand to brand relations, celebrity to brand relations, and clusters of related entities.

2.2 Data Retrieval

The manually determined set of queries used to retrieve unique video IDs included:

- Met Gala 2026 live
- Met Gala 2026 reaction
- Met Gala 2026 review
- Met Gala 2026 looks
- Met Gala 2026 theme

Video IDs and all available metadata for each video were retrieved via the YouTube Data API. The fields collected included video title, description, channel ID, channel name, publication date, category, view count, like count, comment count, and any others available.

Each unique video ID was then used to retrieve all associated comment data for a given video. Both top-level parent comments and available child replies were collected. From querying YouTube with the stated search terms a total of 113 unique video IDs were retrieved, all published between the 3/5/2026 and 9/5/2026.

2.3 Data Fields

The primary fields used to perform analysis included:

- Text
- publishedAt

Text data is used to assess sentiment and tone, identify topics by analysing co-occurrence of different words between different comments, capture celebrity and brand mentions for network analysis, and examine how sentiment changes over time using publishedAt times in conjunction with text.

2.4 Data Limitations

Before collection of data, multiple data sources were planned to be used. X and Reddit were selected as suitable sources but due to expensive API costs or restrictive policies were omitted from the data collection. This introduces a bias as YouTube comments may not be reflective of overall comments across different social media platforms. The queries used for search were manually selected and may introduce sampling bias, with top-performing videos across YouTube likely representing a positive sentiment bias. To assist with this bias, the search API was used to determine video IDs over manual video selection.

Misspelling or using different aliases or nicknames are common within YouTube comments and can damage detection accuracy and therefore, total entity counts, network node weights, or co-mention relationships. Videos with significantly higher comment contributions may skew results towards the overall sentiment for a given video. Finally, co-mentions between different brands and celebrities is not proof of a relationship between the two entities. Context gathered from inspecting the comment manually may indicate that the celebrities were mentioned in an unrelated manner.

2.5 AI Declaration

Publicly available Met Gala 2026 guest lists were scraped in order to effectively detect mentions between different celebrities and designers. Furthermore, detect co-mentions within comments. AI was then used to generate a simple JSON format list of primary names with type and reasonable aliases for the given entity. The combination of name and aliases was then used to detect mentions across different comments allowing for reasonably accurate counts to be determined.

361 different celebrities and 120 different brands were included as potential nodes. Prior to assembling this data structure, mention counts were unusually low. Using this entity mapping allowed for constant time access to a list of designer brands or celebrities and a reliable network to be formed that was more robust to the use of aliases or minor character differences.

```
"maison_margiela": {
  "name": "Maison Margiela",
  "type": "designer_brand",
  "aliases": [
    "maison margiela"
  ]
},
"marc_jacobs": {
  "name": "Marc Jacobs",
  "type": "designer_brand",
  "aliases": [
    "marc jacobs"
  ]
},
```

3. Data Preprocessing

3.1 Data Collection Summary

Comments were deduplicated resulting in the removal of 202 duplicate rows. A random sample of comments was then selected to inspect data quality issues to address in pre-processing. Comments were flattened into single row-per-record form for easier processing.

Data Summary

Metric	Quantity
Video IDs	110
Total Comments	63,250
Unique Comments	63,048
Parent Comments	48,950
Child Replies	14,098
Empty comments	13

3.2 Data Inspection

As shown in the comments, emojis, mentions, irregular case, unicode letters, and comments containing no text characters are contained in the dataset. These comments and other examples were used to determine pre-processing to apply to the dataset.

Random Sample of Comments Inspected For Data Quality Issues

Comment Content (Raw)
👉👉👉👉👉👉👉👉
MISS JENNIE KIM!
@prpowell4038🤔 for the fact she's entitled she might be acting

the same way as North. Get out of gay Zee's and be-yawn 🤔nce's asses
Thankieeee

3.3 Language Detection

For the analysis, only English comments are to be kept to ensure sentiment and topics are interpretable. Python langdetect was used to determine non-English comments

Top 5 Language Detections

Language Key	Count	Distribution (%)
en	46,829	74.28%
unknown	1,679	2.66%
so	1,563	2.48%
pt	1,144	1.81%
de	1,074	1.70%

Non-English comments were inspected to confirm reasonable accuracy and all non -English detected and removed from the flattened comment rows. This resulted in a total of 46,829 (74.3%) comments being kept and 16,219 (25.7%) comments removed.

3.4 Preprocessing Pipeline

Preprocessing Pipeline Cleaning Steps

Analysis Type	Cleaning Steps Applied
Entity Matching	html entities, URLs, timestamps, mentions, letter accents, and stripping to only alphabetic characters (to improve matching with entity names and aliases)
Sentiment Analysis	html entities, URLs, timestamps, mentions, and whitespace normalisation (for compatibility with VADER and RoBERTa)
Topic Analysis	Entity matching cleaning steps + normalisation of unicode, digits, punctuation, and whitespace normalisation

An example of the preprocessing pipeline is shown below. Sets of preprocessing steps where changes did not occur are combined into a single step.

Preprocessing Step-by-Step Example

Preprocessing Step	Example 1: Comment (7/10)	Example 2: Comment (1/10)
Raw Comment	[7/10] Jisoo and Karina look soooo AI They're unreal 🤔	[1/10] OMG Jisoo is so prettyyyy
remove_html_entities	Jisoo and Karina look soooo AI They're unreal 🤔	OMG Jisoo is so prettyyyy

remove_urls, remove_timestamps, remove_mentions	Jisoo and Karina look soooo AI They're unreal 🤖	OMG Jisoo is so prettyyyy
lowercase_and_strip	jisoo and karina look soooo ai they're unreal 🤖	omg jisoo is so prettyyyy
normalise_whitespace	jisoo and karina look soooo ai they're unreal 🤖	omg jisoo is so prettyyyy
remove_unicode	jisoo and karina look soooo ai theyre unreal	omg jisoo is so prettyyyy
remove_digits, strip_punctuation, normalise_whitespace_after_punctuation	jisoo and karina look soooo ai theyre unreal	omg jisoo is so prettyyyy
tokenize	['jisoo', 'and', 'karina', 'look', 'sooo', 'ai', 'theyre', 'unreal']	['omg', 'jisoo', 'is', 'so', 'prettyyy']
remove_stopwords	['jisoo', 'karina', 'look', 'sooo', 'ai', 'theyre', 'unreal']	['omg', 'jisoo', 'prettyyy']
stem_tokens	['jisoo', 'karina', 'look', 'sooo', 'ai', 'theyr', 'unreal']	['omg', 'jisoo', 'prettyyi']

The final set of flattened, preprocessed comments were stored in comments JSON using appropriate keys and dumped to preprocessed JSON to be used for analysis. Comments that were empty post-processing were culled from the comment set.

3.5 Preprocessing Summary

Preprocessing Summary

Process	Before	After	Reduction (%)
Language filtering	63,048	46,808 (English Kept)	25.8%
Sentiment text	4,575,121	4,493,271	1.8%
Topic Text Characters	4,575,121	2,826,266	38.2%
Topic Tokens	971,761	436,276	55.1%
URLs removed	46,808	15 affected	0.03%
Timestamps removed	46,808	1,462 affected	3.12%
Mentions removed	46,808	3,730 affected	7.97%
Empty Rows (Sentiment Text)	46,808	19 affected	0.04%
Empty Rows (Topic Text)	46,808	283 affected	0.60%

The preprocessing pipeline aimed to ensure interpretability and improve text quality. This was achieved by enforcing English as the detected language (constrained by library accuracy), and stripping noise elements like mentions, URLs, and timestamps, and removing comments that became empty post-processing. These steps resulted in significant reductions in character and token counts for text used in topic and network analysis, reducing overall noise and improving the final data quality.

4. Sentiment Analysis

4.1 Goal

Preprocessed comment data will have sentiment analysis used to determine the distribution of positive, negative, and neutral comments across all of the collected data. Analysis will also be performed to identify sentiment towards different celebrities and brands. A final “accurate” comment set will be determined by combining results where VADER and RoBERTa agree on classification. This final set will be used to determine a sentiment distribution with improved accuracy .

4.2 Model Justifications and Limitations

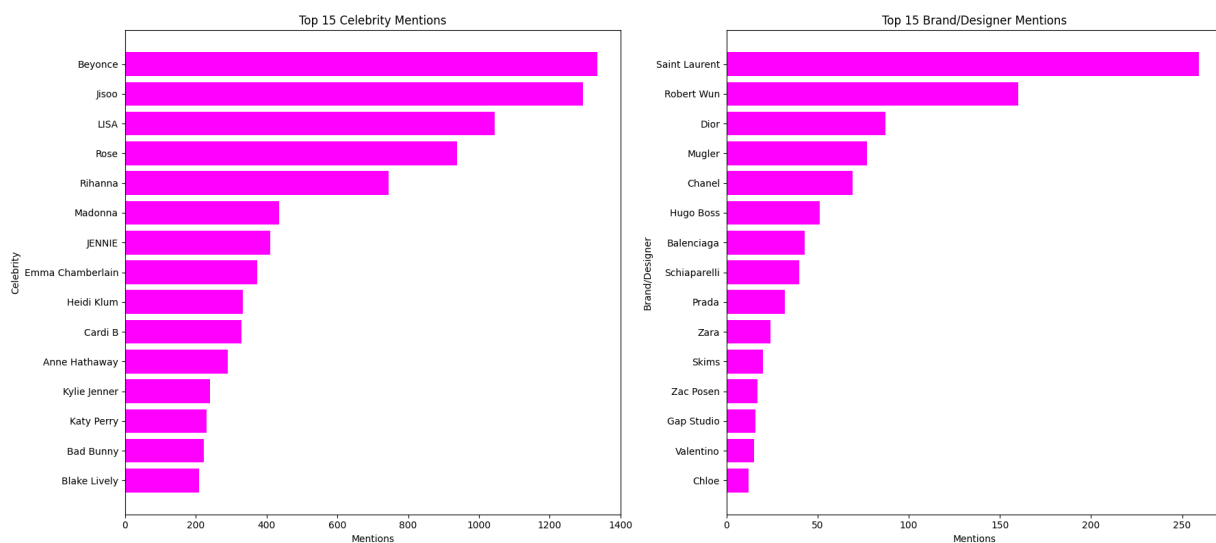
VADER (Valence Aware Dictionary and sEntiment Reasoner) is a lexicon and rule-based sentiment analysis tool designed for analysing social-media text (Hutto & Gilbert, 2014). VADER is trained to handle simple negations, contractions as negations, word-shape (e.g. capitalisation as a representation of stronger emotion), degree modifiers, emojis as signals, and other useful text patterns. This ability to handle more complex text patterns is expected to improve sentiment detection accuracy for YouTube comments, which often have non-simple structure and use emojis and capitalisation to express sentiment.

The second model to be used is a RoBERTa-based (*Robustly Optimized BERT Approach*) model fine-tuned on Twitter tweets (Cardiff NLP, 2026; Liu et al., 2019). The model classifies comments as positive, neutral, negative, or unknown using a defined confidence threshold. If this threshold is exceeded for a given label, then the comment is assigned that label. The chosen variant of RoBERTa is trained and fine-tuned on social media text and capable of identifying context-based language patterns, identifying how words relate to each other across the entire sentence.

The core comment set for analysis will be determined by identifying matching classifications between VADER and RoBERTa, excluding any unknown sentiment results from the latter. Both models primarily struggle with misclassifying sarcasm or irony and remain dependent on the overall quality and interpretability of the comment data.

4.3 Entity Counts

Top 15 Celebrity and Designer Brand Mentions



Celebrity and Designer Brand Mentions

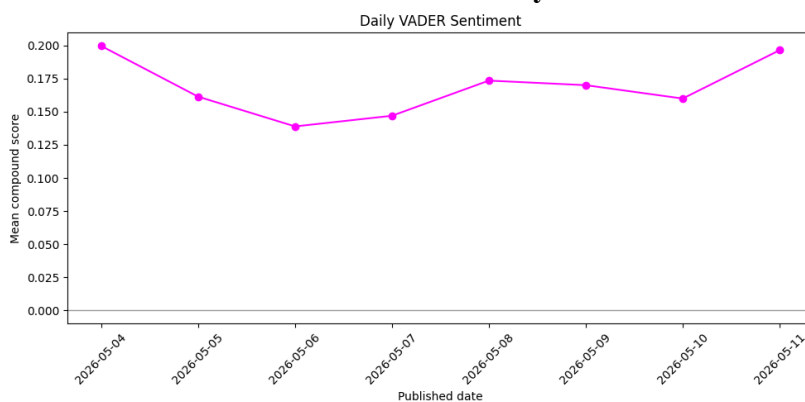
Value	Celebrity	Designer Brand
Total unique	215	66

Total mentions	13,758	1,112
Mentions (Mean)	63.99	16.85
Mentions (Median)	10	4.5
Mentions (Max)	1,334	259

Entity counts were used to determine a reasonable median occurrence count per celebrity or brand. Since celebrities had significantly higher mention counts, applying a consistent cut-off resulted in fewer than 15 designer brands qualifying. The minimum count cut-off was adjusted to 7 for designer brands to ensure 15 total entities could be listed for each entity type for sentiment distribution per entity comparisons. Additionally, the entity counts will be used for later network analysis

4.3 VADER Sentiment Analysis

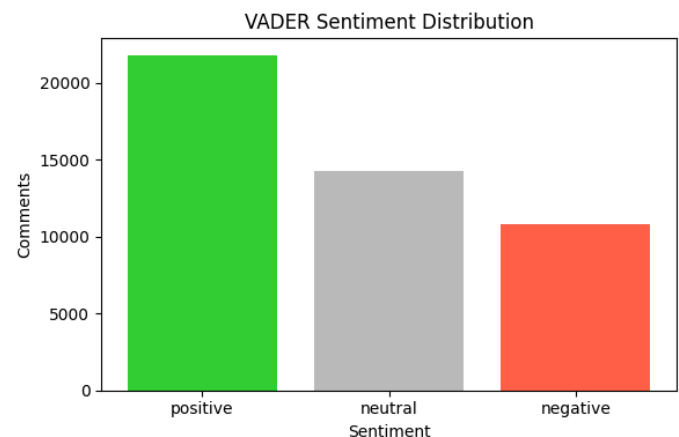
VADER Sentiment Distribution Per Day



VADER sentiment series determined using published dates and average compound score reveals a stable graph. The overall average sentiment is consistently above the positive default threshold of 0.05. VADER indicates a minor dip on the 5/6/26 to 6/6/26 which are the two days following the actual event on the 4/6/26. The plausible reason for this dip is that videos filming reactions and discussions were likely released just after the event on these dates, and if these videos were engineered for engagement, might have encouraged negative comments. For the other sections of sentiment-analysis, daily sentiment distribution will be omitted because the event was a static, single-day occurrence, and sentiment is therefore not expected to change over time.

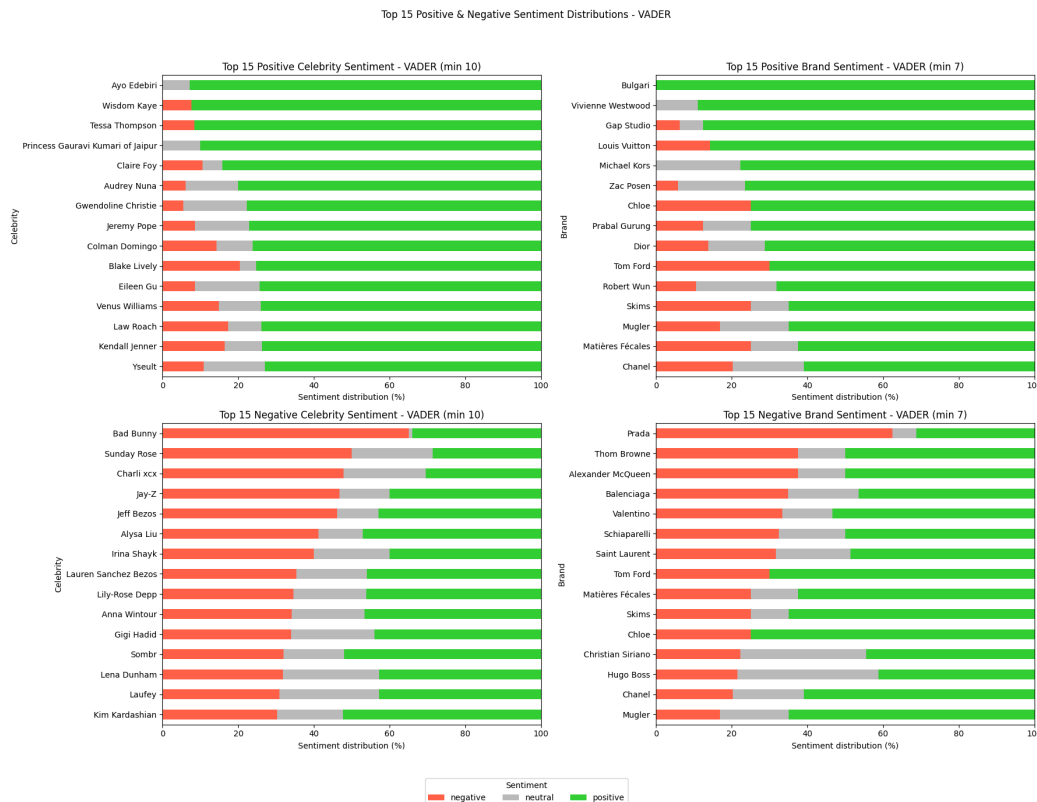
VADER Sentiment Distribution

Label	Count	Distribution
Positive	21,782	46.51%
Neutral	14,273	30.48%
Negative	10,774	23.01%



VADER detected overall positive sentiment representing 46.51% of the processed comment distribution. The high distribution of neutral classifications (30.48%) indicates VADER struggled to assign sentiment to a large proportion of comments. Comparing VADER and BERT sentiment will provide a higher-confidence set of classifications.

Top 15 Positive/Negative Sentiment Distributions (VADER)



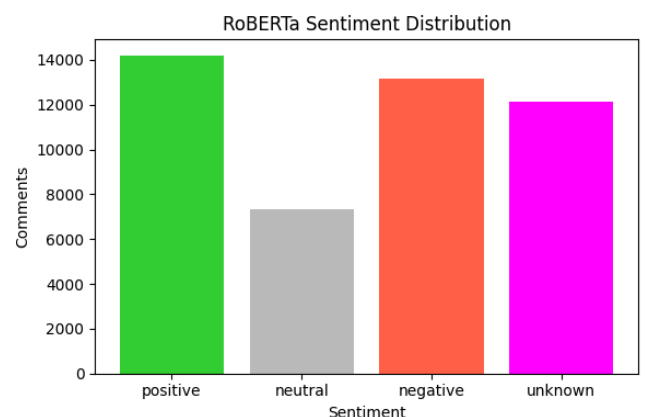
The above graphs reveal the sorted top-15 positive and negative distributions for celebrities and brands. Celebrities use a minimum count cut-off of 10, selected because it reflects the median count across all celebrities, while brands use a cut-off of 7. These thresholds reduce the risk of low-count entities appearing highly positive or negative due to only one or two comments.

As indicated in these graphs, positive sentiment remains dominant, even within the top-15 negative entity distributions. Additionally, many rows show relatively even or mixed distributions across positive and negative sentiment, suggesting that VADER is not accurately able to classify entities as strongly positive or strongly negative. The graphs are useful to roughly assess overall sentiment, but the negative VADER rankings should be interpreted cautiously.

4.4 RoBERTa Sentiment Analysis

RoBERTa Sentiment Distribution

Label	Count	Distribution (%)
Positive	14,198	30.32%
Neutral	7,332	15.66%
Negative	13,161	28.10%
Unknown	12,138	25.92%



RoBERTa's analysis showed a lower overall positive sentiment at 30.32%, contrasting significantly with VADER's 46.51%. The difference stems largely from RoBERTa's ability to classify a large portion of previously 'Neutral' comments (30.48% in VADER) into the 'Unknown' category (25.92%). This suggests that RoBERTa is better at distinguishing between 'Neutral' and 'Unknown'

classifications which lead to a more defined distribution of the three core sentiments, and an increased negative classification (28.10% vs 23.01% in VADER).

Top 15 Positive/Negative Sentiment Distributions (RoBERTa)



These graphs show that RoBERTa sentiment remains strongly positive across both celebrity and brand mentions. The minimum count thresholds help reduce distortion from rarely mentioned entities, making the rankings more reliable. Even among the “most negative” entities, positive sentiment is still present, which supports the wider finding that audience discussion of the 2026 Met Gala was generally favourable rather than predominantly critical. Compared with VADER, RoBERTa produces less variation within each sentiment distribution. Entities classified as positive tend to show strongly positive distributions with only small negative proportions, while entities classified as negative show the opposite pattern. This suggests RoBERTa has higher accuracy with classification, whereas VADER produces more mixed sentiment distributions suggesting poorer accuracy.

4.5 Combined Sentiment Analysis

The core comment set for analysis was determined by identifying matching classifications between VADER and RoBERTa, excluding any unknown sentiment results. Matching classifications between models aimed to validate accuracy using both methods of analysis to build a stronger case for identified sentiment across different comments. The matched results indicate that positive sentiment remains the dominant category. While RoBERTa is more appropriate as the primary model due to its context-based language patterns, the matched classification confirms that an overall positive sentiment can still be seen.

4.6 VADER vs BERT Classifications

VADER	BERT	Comment
neutral	positive	At least the South Asian contingent brought ART to the Fashion Is Art themed gala.
positive	negative	But when other people wear other shapes, you also laugh

VADER	BERT	Comment
		at them, Halloween, ghosts from the future, stiff sticks, four legs boogies... What do you want from them?
neutral	positive	the daughter mother duo ❤️
neutral	negative	also the proportions are sooooo off
negative	neutral	Katy's mirror mask was referencing the mannequin heads in the exhibit, no?

Across this sample of disagreements, RoBERTa can be seen to identify context-based sentiment that VADER is unable to. The poor accuracy in this random sample of VADER misclassifications is common across different samples. Across these instances of disagreement, RoBERTa can be seen to identify context-based sentiment that VADER is unable to capture. Comments such as “brought ART” and “the daughter mother duo ❤️” are better interpreted as positive, while comments referring to “proportions” being “sooooo off” are clearly negative despite VADER assigning a neutral classification. These examples suggest VADER struggles with assigning sentiment to nuanced language that RoBERTa is more attuned at analysing. Therefore, RoBERTa is considered the more reliable sentiment classifier and VADER is used as a secondary sentiment analysis method to verify against.

5. Network Analysis

5.1 Graph Structure & Construction

Our network analysis adopts two graphs;

Celebrity - Celebrity co-mention graph

We build a graph of all celebrities co-mentioned in user comments. Using the number of times the celebrities are co-mentioned with each other as edge weights.

Celebrity - Brand co-mention graph

We build a second graph using pairs of brands and celebrities co-mentioned together.

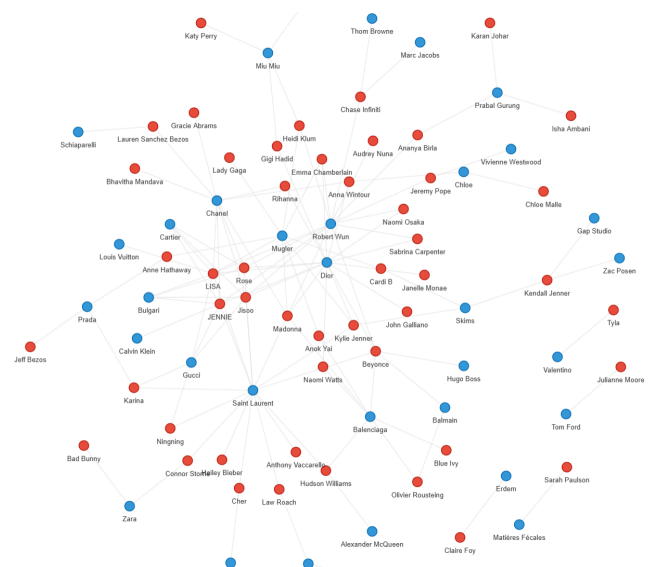
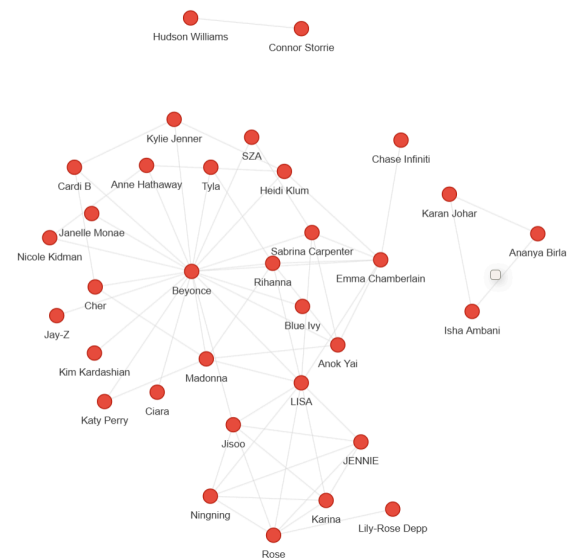
Blue nodes denote a brand whilst red nodes denote a celebrity.

Edges are weighted by the number of times brands and celebrities are co-mentioned.

We chose to go with a celebrity co-mention as we wanted to extract information on the few key aspects of the Met Gala. The celebrities that attend the events, and The brands in support of these celebrities.

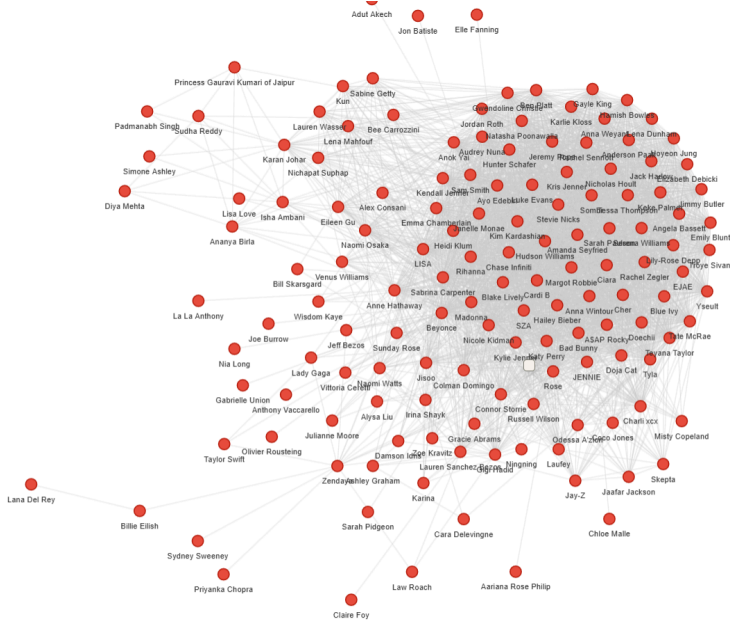
Due to the nature of co-mentions, we adopt an undirected graph, weighted by the number of co-mentions on a specific combination of celeb-celeb or celeb-brand.

The graph weights would also be useful for filtering out less popular nodes, The full celebrity graph alone was too congested with nodes that had very little co-mentions but still received an edge



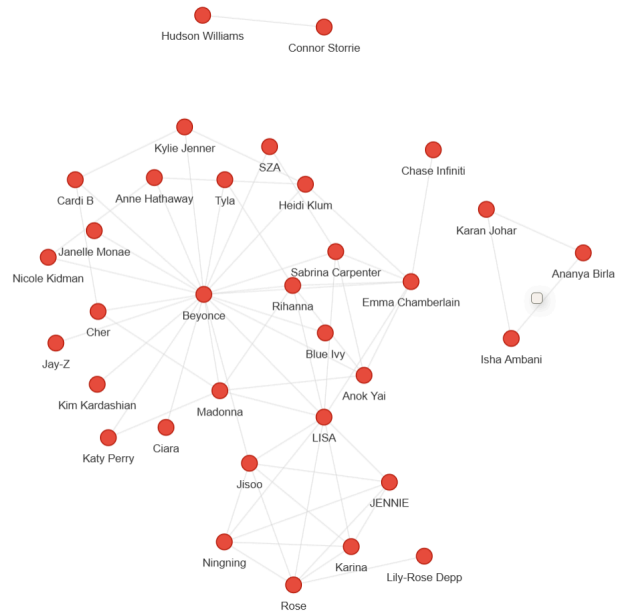
as such we remove edges with less than a set number of co-mentions and remove isolated nodes.

Celebrity Graph PRE-Filtering



Celebrity Graph POST-Filtering

Minimum Co-mentions = 15



Basic Network summary:

Celebrity Graph:

GRAPH SUMMARY	
nodes	33
edges	63
density	0.119318
is_directed	False
is_connected	False
avg_degree	3.8182
node_types	{'celebrity': 33}

Celebrity Brand Graph:

GRAPH SUMMARY	
nodes	83
edges	120
density	0.035263
is_directed	False
is_connected	False
avg_degree	2.8916
node_types	{'celebrity': 51, 'brand': 32}

The basic network summary allows us to infer a few things. A **celebrity on average is co-mentioned with 4 other celebrities**, while **each brand is co-mentioned with roughly 3 celebrities**. The density of both graphs is sparse suggesting **celebrities tend to cluster in specific discussion groups rather than being universally co-mentioned**

5.2 Centrality Analysis

Top 5 results of Centrality for Celebrity Graph is shown below;

Degree	Betweenness	Closeness	Eigenvector
1. Beyonce (0.656)	1. Beyonce (0.495)	1. Beyonce (0.670)	1. Beyonce (0.493)
2. LISA (0.313)	2. LISA (0.150)	2. LISA (0.518)	2. LISA (0.365)
3. Emma Chamberlain (0.219)	3. Jisoo (0.069)	3. Emma Chamberlain (0.475)	3. Emma Chamberlain (0.268)
4. Rihanna (0.188)	4. Emma Chamberlain (0.062)	4. Rihanna (0.465)	4. Rihanna (0.260)
5. Jisoo (0.188)	5. Rose (0.052)	5. Jisoo (0.465)	5. Madonna (0.242)

Top 5 results of Centrality for Celebrity - Brand Graph is shown below;

Degree	Betweenness	Closeness	Eigenvector
1. Dior (0.195)	1. Saint Laurent (0.299)	1. LISA (0.363)	1. Dior (0.361)
2. Saint Laurent (0.183)	2. Robert Wun (0.254)	2. Madonna (0.359)	2. Robert Wun (0.317)
3. Robert Wun (0.171)	3. Dior (0.202)	3. Saint Laurent (0.346)	3. Saint Laurent (0.313)
4. Chanel (0.122)	4. Mugler (0.112)	4. JENNIE (0.344)	4. LISA (0.291)
5. LISA (0.110)	5. Chanel (0.111)	5. Jisoo (0.341)	5. JENNIE (0.268)

A few notable insights from the above data;

Beyonce dominates every single measure; degree (0.656), betweenness (0.495), closeness (0.670), eigenvector (0.493). This means she is not just the most mentioned celebrity but the structural centre of the entire discussion network. She is the node a majority of other nodes orbit.

LISA ranks 2nd across all four measures. Although her mention count (1,045) is lower than Jisoo (1,299). She still consistently outranks Jisoo on centrality. This tells us LISA is not just frequently mentioned; she is mentioned alongside a more diverse range of other celebrities, making her a broader cultural connector than her raw mention count suggests.

Emma Chamberlain Ranks 3rd across all four metrics, with a fairly low mention count (367). This suggests although Emma is not mentioned often, she is mentioned with highly central celebrities.

Rose makes the top 5 list in only the betweenness centrality; this suggests rose is not a central celebrity however, rose acts as a bridge connecting different celebrity clusters to one another.

Other highly central celebrities include **Jisoo, Rhiana, Madonna**.

In the Celebrity Brand Graph;

Naturally the top degree centrality nodes are brands; **Dior, Saint Laurent, Robert Wun, Chanel**. Each brand is co-mentioned with a number of celebrities.

Robert Wun stands out in the list; despite only having 161 mentions RW ranks 2nd in betweenness centrality and eigenvector centrality. This suggests that RW is niche yet still central to the brand celebrity structure. RW has probably designed dresses for highly central celebrities.

Saint Laurent ranks top in betweenness centrality, suggesting SL despite not being the most mentioned brand (257) is closely co-mentioned with a large number of celebrities.

LISA & JENNIE (Both BlackPink members) appear mostly alongside luxury brands. Highlighting the Black Pink - Luxury fashion trend.

5.3 Community Detection Analysis

Celebrity Graph Community Detection

Algorithm	Modularity (Q)	Quality
Louvain	0.4707	Good structure connections
Greedy	0.4707	Good structure connections

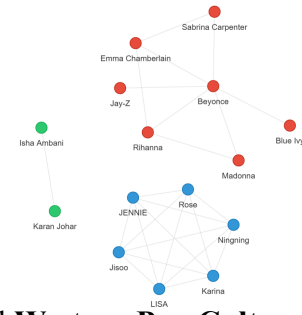
Celebrity Brand Graph Community Detection

Algorithm	Modularity (Q)	Quality
Louvain	0.5267	Very good structure connections
Greedy (Chosen)	0.5134	Very good structure connections

The results of this are shown below;

Celebrity graph communities;

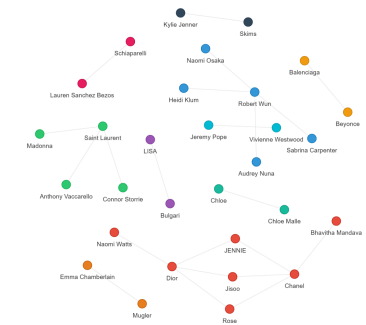
Community ID	Node Count	Top Members
Community 0	7	Beyonce, Rihanna, Madonna, Emma Chamberlain...
Community 1	6	Jisoo, LISA, Rose, JENNIE, Ningning...
Community 2	2	Karan Johar, Isha Ambani...



- **Community 0: Beyonce, Rihanna, Madonna, Emma.** Are all **Western Pop Culture** icons. From the centrality analysis above and graphs visual we can infer that Beyonce and Rihanna are the central nodes of this cluster with Madonna, Emma and Sabrina being notable figures in the Western Pop Culture sphere.
- **Community 1: Jisoo, LISA, Rose, JENNIE, Ningning.** Are all members of **Black Pink or Korean Pop community**. We can infer with Rose's betweenness centrality score that she probably acts as the bridge connecting this community to community 0 western pop culture.
- **Community 2: Karan Johar, Isha Ambani.** Are both prominent Indian celebrity figures. Karan Johar is a Bollywood actor, whilst Isha Ambani is a popular business figure. This community reflects Indian attendees at the MetGala

Brand graph communities;

Community ID	Node Count	Top Members
Community 0	7	Jisoo, Rose, JENNIE, Dior, Chanel...
Community 1	5	Heidi Klum, Sabrina Carpenter, Robert Wun, Audrey Nuna...
Community 2	4	Madonna, Saint Laurent, Connor Storrie, Anthony Vaccarello...



- **Community 0: Jisoo, Rose, JENNIE, Dior, Chanel.** 5 celebrities all BLACKPINK/KPOP members. This reflects brand ambassador relationship of these celebrities to luxury brands e.g. (Jennie is Chanel's ambassador / Jisoo is Dior's ambassador)
- **Community 1: Heidi, Sabrina, Robert Wun, Audry, Naomi**
- **Community 2: Madonna, Saint Laurent, Connor, Anthony.** This reflects celebrities in the Saint Laurent sphere. Anthony Vaccarello is Saint Laurent's creative director and other celebrities are closely tied to the brand Saint Laurent.
- **Community 3 - 9:** Are all celebrities and their brand sponsors; Beyonce-Belenciaga, LISA-Bulgari, Chloe-Chloe Malle, Emma-Mgler...

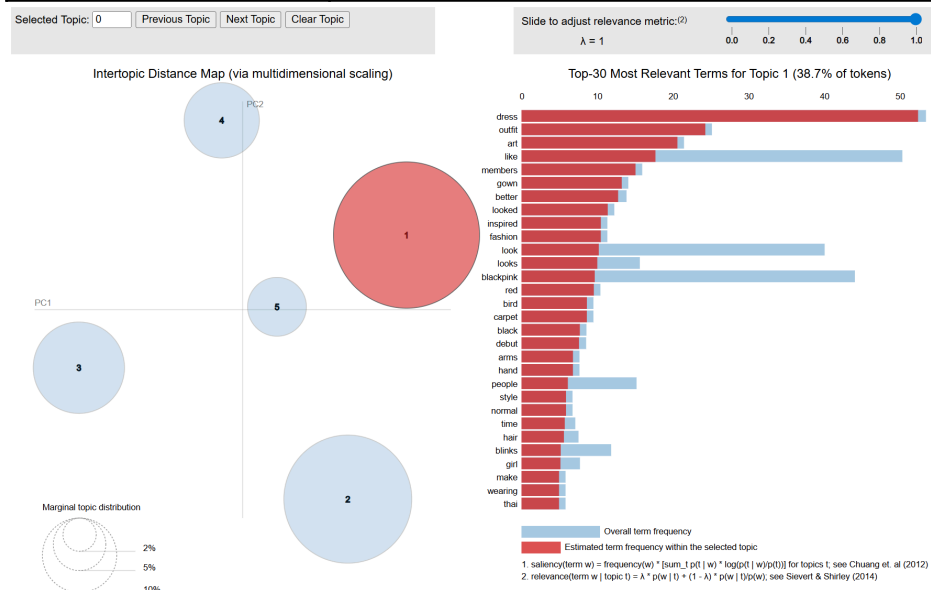
5.3.1 Topic Analysis on Communities

We also run Latent Dirichlet Allocation (LDA) on communities in our celebrity graph to try to understand topics discussed in different communities.

Community 1 - KPOP Community

Topic	Keywords
Topic 1: General Met Gala Comments	gala like met really come know theme good beautiful look slayed wore looking amazing comes dresses new plain
Topic 2: BLACKPINK INTERVIEW	blackpink interview love blinks girls blink stunning mermaid giving hard think year members beautiful amazing
Topic 3: KPOP BRAND ASSOCIATIONS	year best mean creative ysl event think dior hair girls literally blinks giving debut better dress good ate stunning plain

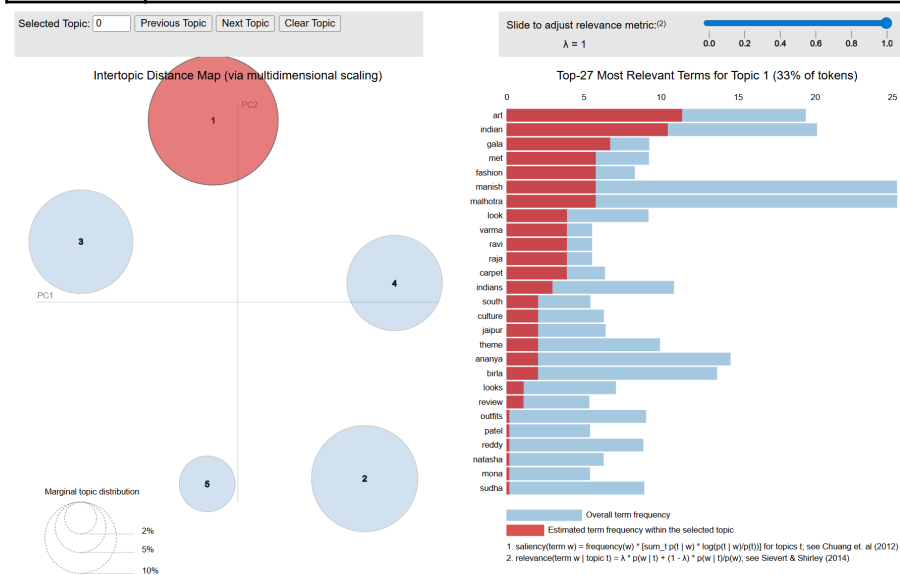
Topic 4: BLACKPINK FASHION CRITIQUE	dress outfit art like members gown better looked inspired fashion look looks blackpink red bird carpet black debut
Topic 5: NINGNING FASHION CRITIQUE	ningning ate pretty look boring outfits karina looks creative hard nailed girl stunning mermaid time pink better dress



Community 2 - Indian Pop Culture Celebrities

All 5 topics relates to Indian Culture and Fashion

Topic	Keywords
Topic 1	art indian gala met fashion malhotra manish look raja ravi varma carpet indians south culture jaipur theme ananya
Topic 2	malhotra manish looks indian theme ananya review culture natasha reddy indians sudha birla carpet gala met patel
Topic 3	outfits art indian birla ananya indians theme met fashion gala south sudha natasha reddy malhotra manish look
Topic 4	manish malhotra birla ananya mona patel look jaipur sudha natasha reddy art raja ravi varma review south gala
Topic 5	reddy manish malhotra sudha south indians mona patel outfits birla fashion jaipur review natasha ananya indian



6. Topic Modelling & Analysis

6.1 Approach: LDA vs NMF

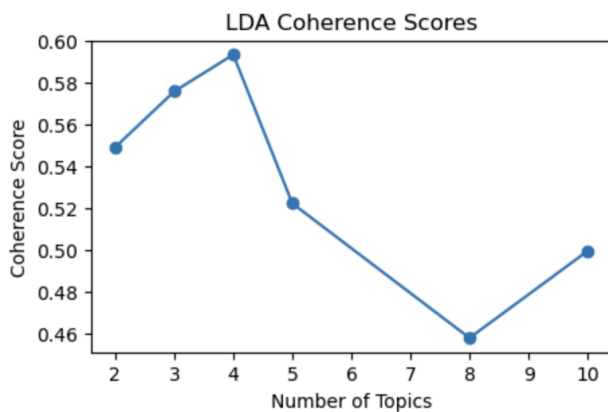
To explore latent themes in the collected Youtube comments, two topic modelling approaches were applied: Latent Dirichlet Allocation (LDA) and Non-negative Matrix Factorisation (NMF).

Preprocessing and Feature Engineering

The text data was further preprocessed by removing standard and domain-specific stopwords to remove noise from conversational language (e.g. “love”, “really”, “video”). To improve semantic capture, both unigrams and bigrams were used, allowing the model to identify meaningful phrases important to fashion-related discourse.

LDA Topic Modelling

LDA was first applied as a baseline probabilistic topic model, where each document is assumed to be a mixture of topics.



To determine the optimal number of topics, multiple LDA models were trained and evaluated using topic coherence scores.

As shown in the figure, models with 2 to 4 topics achieved the highest coherence scores, indicating better semantic consistency among top words within topics. Based on this, the best-performing configurations were further examined qualitatively by reviewing example comments assigned to each topic, in order to assess how well the topics aligned with actual data and whether they were

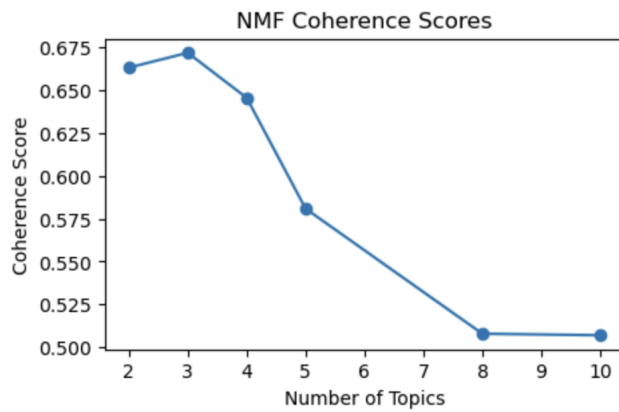
meaningfully coherent.

While LDA was able to identify broad thematic structures within the data, several limitations were observed, as also displayed above. In particular, there was noticeable overlap between topics with similar keywords appearing across multiple themes (e.g. “dress”, “outfit”, “beyonce”). Additionally, some topics contained generic conversational terms that reduced interpretability. These issues were further amplified by the short and informal nature of the comments, which made it more difficult for the model to form clearly separated and stable topic distributions.

```
=====
TRAINING LDA (4 TOPICS)
=====
Coherence Score: 0.5936
TOPIC 0
-----
theme, art, fashion, outfit, boring, outfits, wear, dress, fashion_art, event
TOPIC 1
-----
beyonce, shes, need, world, money, real, ridiculous, want, right, women
TOPIC 2
-----
dress, jisoo, lisa, blackpink, emma, stunning, rose, emma_chamberlain, ate, jennie
TOPIC 3
-----
beyonce, looks_like, rihanna, thought, looked, outfit, thank, face, giving, thats
```

NMF Topic Modelling

In light of these limitations, Non-Negative Matrix Factorisation (NMF) was applied as an alternative topic modelling approach, using TF-IDF weighted features instead of raw counts. Unlike LDA, NMF is based on matrix factorisation rather than a probabilistic framework, and tends to work better on short and sparse text data such as social media comments (Gillis 2014).



Similar to LDA, to select the optimal number of topics, several NMF models were trained and compared using topic coherence scores. As shown in the figure, the 3-topic model achieved the highest coherence score, followed closely by the 2-topic and 4-topic models. These models were further inspected by examining the comments associated with each topic to determine how well the discovered themes aligned with the underlying discussions in the data.

Although the 3-topic model achieved the highest coherence score, the 4-topic model was ultimately selected as it provided better thematic separation and more interpretable topic groupings. In particular, the additional topic allowed fandom-driven discussions centered around specific celebrities and groups to be separated from broader outfit and fashion commentary, resulting in clearer distinctions between themes.

```
=====
TRAINING NMF (4 TOPICS)
=====
Coherence Score: 0.6455

TOPIC 0
-----
theme, art, fashion_art, fashion, costume_art, boring, outfit, literally, outfits, rose

TOPIC 1
-----
dress, art, looks_like, rose, lisa, black, fashion, wearing, gorgeous, boring

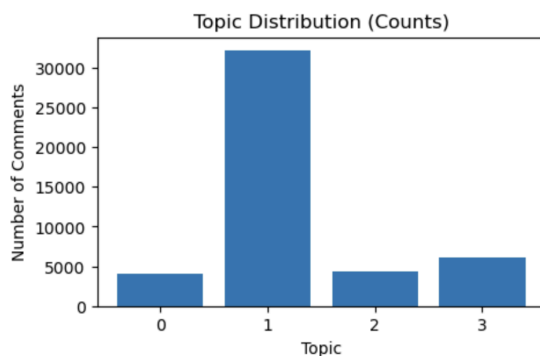
TOPIC 2
-----
jisoo, lisa, blackpink, outfit, finally, rose, queen, jennie, ate, stunning

TOPIC 3
-----
beyonce, queen, rihanna, outfit, looked, skeleton, looks_like, wearing, sorry, loved
```

Based on both coherence scores and qualitative inspection, NMF was selected as the final topic modelling approach for subsequent analysis.

6.2 Topic Analysis

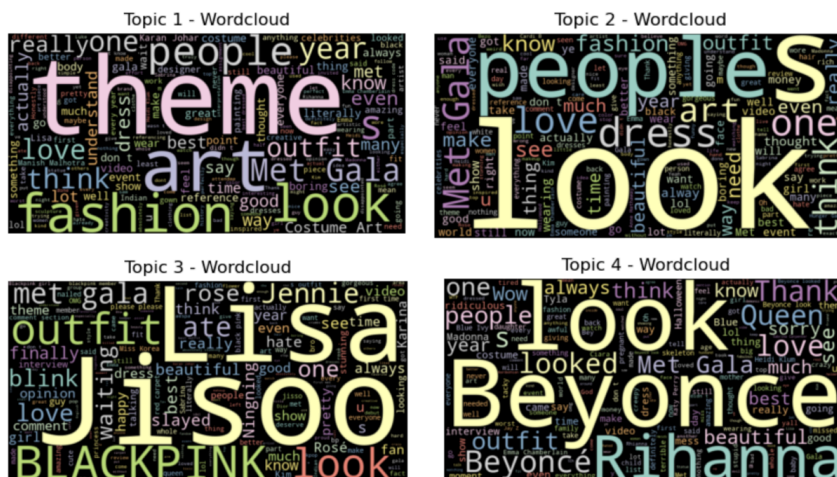
6.2.1 Topic Distribution



The topic distribution shows that one topic clearly dominates the discussion, containing 32,205 comments (68.80%) of the dataset. The remaining three topics are much smaller in comparison, each representing under 15% of all comments (8.56%, 9.44% and 13.20% respectively), which suggests that while there are several distinct strands of conversation, most users are clustered around a single, broad area of discussion. This imbalance is important to keep in mind when interpreting the subsequent topic-wise sentiment analysis.

6.2.2 Topic-Wise Analysis

Each topic was visualised using wordcloud visualisations, as shown in the figure below:

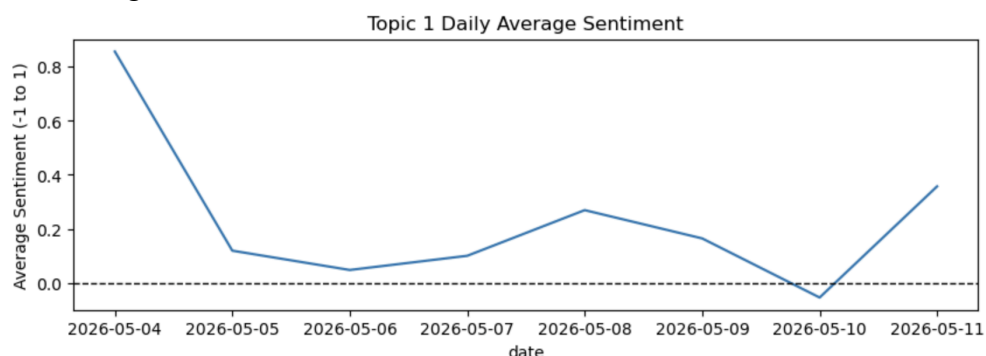


These visualisations show the most common terms appearing in each topic, revealing the main themes present in the dataset. Based on these recurring keywords, the topics were interpreted and labelled as follows:

- **Topic 1 - Discourse about Met Gala Theme: "Fashion is Art"**

This topic focuses on how online audiences debated on whether the Met Gala theme “Fashion is Art” was successful or not.

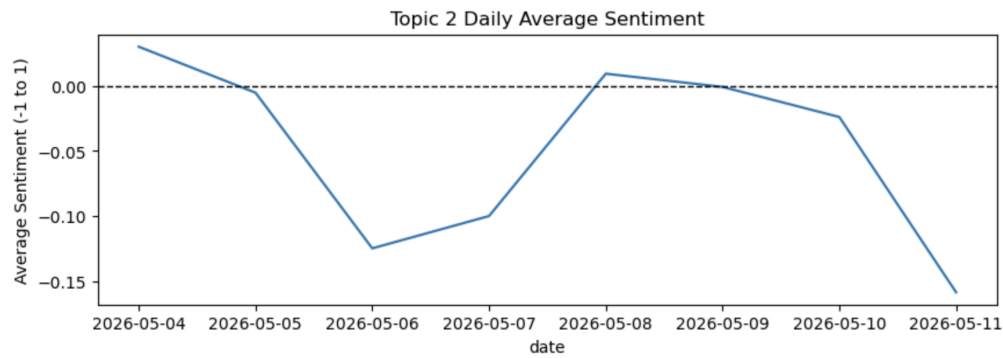
As shown in the below figure, sentiment is strongly positive on 4 May, the day of the event, suggesting an enthusiastic initial response. In the days that follow, the average sentiment declines to a mildly positive level, with a noticeable dip around 10 May, which may reflect later critiques and social media debates once audiences had time to re-watch, compare, and judge the outfits against the theme.



A positive example comment is: “*This year's met gala theme was so nice (so are the dresses!!)*” highlighting approval of both the concept and styling. In contrast, the negative comment, “*Poor tasted costume party ACK,!! WHOM IN THE UNDERGROUND WORLD CAME UP WITH THE, WHAT? IS THE THEME...* 🙄🤔🤔🤔🤔🤔”, dismisses the event as poorly conceived and executed. Overall, the figure and comments together indicate that while initial reactions were highly favourable, ongoing discussion around “fashion as art” became more polarised over time.

- **Topic 2 - General Outfit and Dress Reactions**

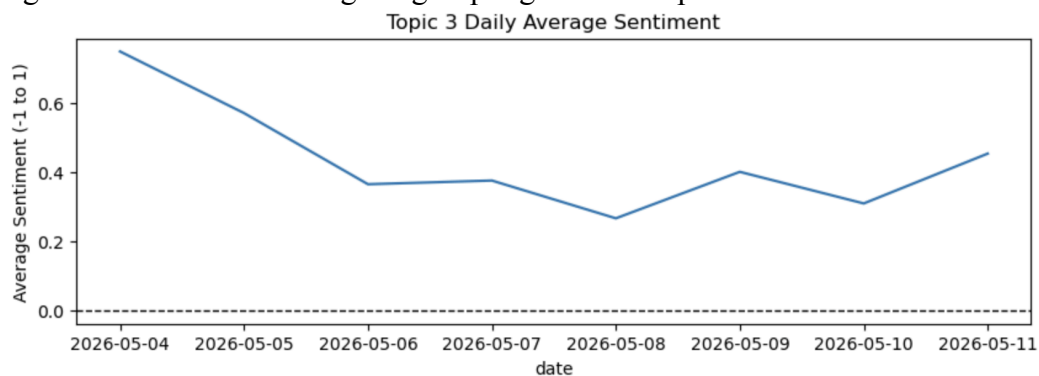
This topic captures audience reactions to the Met Gala outfits overall, focusing on whether the looks were exciting, flattering or underwhelming. The daily average sentiment in the below figure stays close to neutral and dips slightly negative on several days, suggesting that reactions to the fashion were quite divided rather than overwhelmingly positive.



Supportive comments like, “*I am SO excited for this night with y’all 😊🔥 Rob, I love you 🥰 You’re totally my favorite and you look so fab as do all of you!*” show fans celebrating specific guests and their styling. However, this is countered by more dismissive views such as, “*This Met was so LAME!!!!!! 95% of the outfits sucked a***”, which frames the event’s fashion as a collective disappointment. Overall, Topic 2 reflects a tension between excitement for individual looks and broader dissatisfaction with the outfits as a whole.

- **Topic 3 - BLACKPINK fan discourse & band members-focused reactions**

This topic captures fan reactions to BLACKPINK’s appearance at the Met Gala, including excitement about seeing the group together and opinions on each member’s outfit.



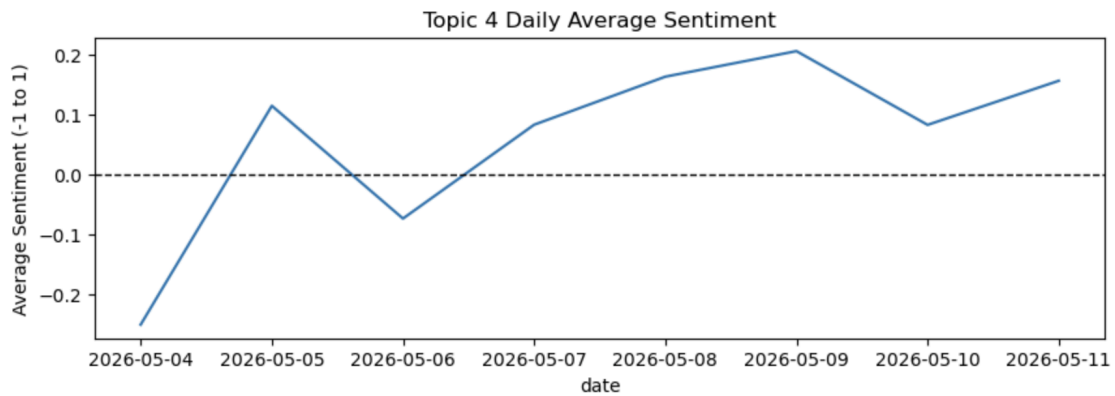
The above figure shows that average sentiment remains consistently positive throughout the week, starting very high on 4 May and staying well above zero despite a slight decline over time. This suggests that fan enthusiasm was strong and sustained, even as more critical views emerged.

A positive example comment is: “*Omg I am very happy to see all the members of blackpink in the met gala 2026. ❤️❤️❤️❤️❤️❤️❤️*”, which emphasises joy at the group’s presence rather than detailed fashion critique. In contrast, a negative comment such as “*in my opinion the worst outfit was lisa’s. i hated it so much.*” shows that within this generally supportive fan space, there was still debate and disagreement about specific styling choices. Overall, Topic 3 reflects highly favourable fan sentiment with instances of intra-fandom criticism directed at particular looks.

- **Topic 4 - Specific Celebrity Red Carpet Looks Reactions**

This topic captures reactions directed at particular celebrities’ Met Gala appearances, especially high-profile figures such as Beyoncé and Rihanna, rather than the event or theme overall.

The below figure shows that daily average sentiment fluctuates around neutral but trends mildly positive over time. There was an initial negative spike on 4 May, followed by a gradual rise above zero, suggesting that early criticism of some looks was later balanced or outweighed by more approving comments as additional photos and perspectives circulated.



Supportive responses such as, “*Thank you to see my girls in MetGala, MetGala2026 is wonderful* 🖤💖” focus on appreciation and excitement at seeing favourite celebrities on the carpet. In contrast, strongly worded critiques like, “*beyonce looks awful i cant believe it. just stay home, ma’am.*” show how individual outfits could trigger harsh backlash. Overall, Topic 4 highlights how celebrity status amplifies both praise and criticism, producing a dynamic but slightly positive stream of reactions to specific red carpet looks.

7. Conclusion

This report examined YouTube audience responses to the 2026 Met Gala to understand who dominated discussion, how they were connected, and what kinds of sentiments and topics emerged. Network and entity analyses showed that a small set of high-profile celebrities and designer brands concentrated most of the attention, and that these figures were frequently co-mentioned in tightly connected clusters, indicating clear communities of related celebrities and brand collaborations.

Sentiment analysis using VADER and RoBERTa, combined with topic modelling, revealed that while overall reactions were mildly positive, discourse was highly polarised. General outfit evaluations dominated the discussions, with additional topics focused on debates about the official theme and fan-centred reactions to specific red carpet looks and people. Together, these findings answer the research questions by demonstrating not only which figures shaped the conversation and how they were associated, but also how audiences emotionally and thematically engaged with the 2026 Met Gala.

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